

Review that CanvasNet Fonts Subsystem Implementation is Correct

2026-10-02

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1. Introduction

1.1 Purpose

This document records the formal review of a set of project files.

1.2 Scope

This review covers the CanvasNet `Fonts` subsystem at the **subsystem level only**. The `Fonts` subsystem comprises two units, `TrueTypeFont` and `SystemFontCatalog`, each of which has its own separate unit-level review-set (`CanvasNet-Fonts-TrueTypeFont`, `CanvasNet-Fonts-SystemFontCatalog`). Per the hierarchical-scope principle in `software-items.md`, unit-level implementation detail is excluded from this review; this review instead relies on the subsystem design document's description of what the two units provide, and verifies that the subsystem's own requirements, design, verification-design, bundled third-party asset documentation, and shared test infrastructure are internally correct, mutually consistent, and consistent with the surrounding system-level context.

Files under review (per `.reviewmark.yaml` review-set `CanvasNet-Fonts`):

- `docs/reqstream/canvas-net/fonts.yaml` (subsystem requirements)

Review that CanvasNet Fonts Subsystem Implementation is Correct

- `docs/design/canvas-net/fonts.md` (subsystem design)
- `docs/verification/canvas-net/fonts.md` (subsystem verification design)
- `test/DemaConsulting.CanvasNet.Tests/TestSupport/SyntheticFontBuilder.cs` (shared test fixture helper)
- `src/DemaConsulting.CanvasNet/Fonts/BundledFonts/README.md` (bundled third-party font licensing/provenance)

Context (not under review, used to understand intended role/scope only):

- `docs/design/canvas-net.md` (system-level design)
- `docs/design/introduction.md` (document conventions)
- `docs/reqstream/canvas-net.yaml` (system-level requirements)

This is a re-review, performed from scratch against `feature/pdf-document-support` commit `bc86abc`, superseding the first formal review of this review-set (performed earlier the same day against an older commit). All files were re-read fresh and all tooling (`reviewmark`, `reqstream`, `markdownlint-cli2`, `cspell`) was re-run against the current state; no conclusions were carried forward from the prior review without re-verification.

1.3 Outcomes

Each check must be recorded with one of the following outcomes:

Outcome	Meaning
Pass	The check was performed and the criterion is satisfied
Fail	The check was performed and the criterion is not satisfied
N/A	The check does not apply; justification is required

1.4 Review Details

Field	Value
Project	CanvasNet
Review ID	CanvasNet-Fonts
Review Title	Review that CanvasNet Fonts Subsystem Implementation is Correct
Fingerprint	6617605dbe8147362457a352881ee020fe723b5882158d303da2727efedd3178
Review Date	2026-10-02 (re-review; commit <code>bc86abc</code> , supersedes the same-day first review)

1.5 Reviewers

Name	Role	Organization	Signature	Date
Formal Review Agent (GitHub Copilot CLI)	Automated Reviewer	DEMA Consulting	(automated)	2026-10-02

1.6 Files Under Review

File
docs/reqstream/canvas-net/fonts.yaml
docs/design/canvas-net/fonts.md
docs/verification/canvas-net/fonts.md
test/DemaConsulting.CanvasNet.Tests/TestSupport/SyntheticFontBuilder.cs
src/DemaConsulting.CanvasNet/Fonts/BundledFonts/README.md

2. Review Checklist

2.1 Requirements Checks

Applicable: Yes

#	Check	Outcome	Justification
REQ-01	All requirements have a unique identifier	Pass	Confirmed via <code>dotnet reqstream --lint</code> (clean) and manual inspection of <code>fonts.ya</code>
REQ-02	All requirements are unambiguous	Pass	Requirement text is specific and scoped per capability.
REQ-03	All requirements are testable	Pass	Each requirement lists a <code>tests:</code> entry naming concrete automated tests, all of which were verified to exist (see REQ-06).
REQ-04	All requirements are consistent	Pass	No contradictions found between sibling requirements.
REQ-05	All requirements are complete (no TBDs)	Pass	No TBD/placeholder markers found.
REQ-06	All requirements are verifiable	Pass	Fixed since prior review. <code>CanvasNet-Fonts-TtcSupport</code> 's <code>tests:</code> list in <code>fonts.ya</code> cites only <code>TrueTypeFont_RealTtcContainer_FaceOne_RendersCffGlyphOutlineAsVisibleInk</code> and <code>CanvasNet_SystemIntegration_LoadFaceFromTtcContainerByIndex_ReturnsExpectedData</code> , both of which exist and pass (confirmed via <code>grep</code> and fresh <code>reqstream --matrix :3/3</code> per test, 3 executions each, no orphan). The previously-cited non-existent test was removed from this requirement.
REQ-07	No compound requirements are present	Pass	Each requirement expresses a single testable capability.
REQ-08	No requirements are missing	Pass	Fixed since prior review. <code>CanvasNet-Fonts-CffOutlineDecoding</code> 's <code>children:</code> list includes <code>CffSeacComposition</code> , <code>CffCharsetResolution</code> , <code>CffWidthResolution</code> , and <code>LoadType1C</code> (previously missing/orphaned). <code>dotnet reqstream --enforce (root-tag=system,quality)</code> reports zero orphaned requirements anywhere in the Fonts area.

2.2 Design Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DES-01	Design documentation clearly describes the purpose of the component	Pass	<code>fonts.md</code> 's Purpose section clearly states the subsystem's responsibility.
DES-02	Design documentation covers the necessary implementation details	Pass	Appropriate subsystem-level detail for both units without unit-internal detail.
DES-03	Design documentation describes how the code is interfaced	Pass	Units/Dependencies/Callers sections document both <code>TrueTypeFont</code> and <code>SystemFontCatalog</code> public surfaces.
DES-04	Design documentation describes the expected normal operation	Pass	Purpose/Units sections describe the normal font-loading/matching flow.
DES-05	Design documentation describes the expected error handling	N/A	Consistent with the repository's established subsystem-doc convention (<code>drawing.md</code> , <code>geometry.md</code>), subsystem-level docs defer error-handling detail to unit-level design documents.

2.3 Technical Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DOC-01	Documentation is free of technical inaccuracies	Fail	<code>docs/verification/canvas-net/fonts.md</code> still cites a non-existent test, <code>TrueTypeFont_RealTtcContainer_FaceZero_RendersGlyphGlyphOutlineAsVisible</code> in three places (Test Environment fixture list, Acceptance Criteria's "seven real-fixture integration tests" count, and the <code>CanvasNet-Fonts-TtcSupport</code> Test Scenarios entry); the test does not exist anywhere in <code>test/</code> (confirmed by exhaustive grep); only 6 tests actually exist in <code>TrueTypeFontRealFontIntegrationTests.cs</code> , not 7. See Finding F1 (remaining).
DOC-02	Documentation is consistent with the current implementation and requirements	Fail	Same issue as DOC-01: the verification doc is now inconsistent with <code>fonts.yaml</code> 's own (now-corrected) <code>tests:</code> list for <code>CanvasNet-Fonts-TtcSupport</code> , which no longer contains the phantom test — the verification doc was not updated to match.
DOC-03	All referenced external documents and dependencies are correctly identified	Pass	<code>BundledFonts/README.md</code> 's Liberation Fonts v2.1.5 (SIL OFL 1.1) and Noto Project substitute font claims re-verified unchanged and accurate; the <code>BundledFonts/</code> folder was not touched by the documentation-sync pass.
DOC-04	Documentation is free of spelling and grammar errors	Pass	<code>markdownlint-cli2</code> and <code>cspell</code> re-run clean against all in-scope Markdown/YAML.

2.4 Code Checks

Applicable: Yes (limited to the one source file in scope: the test-support helper)

#	Check	Outcome	Justification
CODE-01	Code conforms to the project coding standards and style guide	Pass	<code>SyntheticFontBuilder.cs</code> is unchanged since the prior review (confirmed via <code>git diff</code>); re-affirmed Pass.
CODE-02	No obvious resource leaks are present	Pass	No unmanaged resources.
CODE-03	No hardcoded values are present that should be configurable	N/A	Test-fixture builder; hardcoded values are spec-mandated binary layout constants.
CODE-04	Each unit or function has a single, well-defined responsibility	Pass	Unchanged from prior review's assessment.
CODE-05	Code is written at the appropriate level of abstraction	Pass	Unchanged.
CODE-06	Code has an appropriate amount of extensibility for its context	Pass	Unchanged.

2.5 Logic Error Checks

Applicable: Yes (`SyntheticFontBuilder.cs` only)

#	Check	Outcome	Justification
LOGIC-01	Code does only what is intended	Pass	Unchanged from prior review.
LOGIC-02	All significant inputs and boundary conditions are handled correctly	Pass	Unchanged.
LOGIC-03	Concurrency and threading concerns are identified and addressed	N/A	Each instance independent; no concurrency concern.

2.6 Error Handling & Logging Checks

Applicable: No

Not applicable: no production error-handling/logging logic is present in the files under review.

2.7 Usability / Accessibility Checks

Applicable: Yes (documentation usability only)

#	Check	Outcome	Justification
USE-01	The feature or API is easy to use correctly	Pass	Unchanged.
USE-02	All public APIs are well documented	Pass	Unchanged.

2.8 Test Checks

Applicable: Yes (`SyntheticFontBuilder.cs` as shared test infrastructure)

#	Check	Outcome	Justification
TEST-01	Tests cover expected (happy-path) behavior	N/A	Test infrastructure, not a test suite itself.
TEST-02	Tests cover error conditions and boundary cases	N/A	Same reasoning.
TEST-03	Tests are independent and repeatable	Pass	Unchanged: no shared mutable state.
TEST-04	Test names clearly describe the behavior being verified	N/A	Builder methods, not test methods.
TEST-05	New test cases are added for new functionality or defect fixes	N/A	Not assessable from the helper file alone.

2.9 Security Checks

Applicable: No

Not applicable: no authentication, authorization, or untrusted-input handling in scope.

2.10 Code Readability Checks

Applicable: Yes (`SyntheticFontBuilder.cs` only)

#	Check	Outcome	Justification
READ-01	Code is easy to understand	Pass	Unchanged.
READ-02	Methods and functions are small enough to be easily understood	Pass	Unchanged.
READ-03	Symbols are well named	Pass	Unchanged.
READ-04	Code is located in the correct place in the codebase	Pass	Unchanged.
READ-05	Flow of control can be easily followed	Pass	Unchanged.
READ-06	Data flow is understandable	Pass	Unchanged.
READ-07	Comments are provided where the code is non-obvious	Pass	Unchanged.
READ-08	No debug artifacts or commented-out code have been left in the codebase	Pass	Unchanged.

2.11 Requirements vs Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
REQDOC-01	All reviewed requirements are addressed in the general technical documentation	Pass	<code>fonts.md</code> (design) and <code>fonts.md</code> (verification) address all requirement groups in <code>fonts.yaml</code> .
REQDOC-02	No reviewed requirement is contradicted by the general technical documentation	Pass	Fixed since prior review. The parent system-level design doc (<code>docs/design/canvas-net.md</code> , context) now fully documents <code>SystemFontCatalog</code> as the Fonts subsystem's second public unit, including its dependencies (<code>RuntimeInformation</code> , embedded <code>BundledFonts</code>) and external interface table (<code>Fonts</code> , <code>FindBestMatch</code> , <code>LoadBundledFallback</code>) — no longer contradicts the in-scope subsystem design/requirements by omission.

2.12 Requirements vs Implementation Checks

Applicable: No

Not applicable: per the hierarchical-scope principle, unit-level implementation is excluded from this subsystem-level review-set.

2.13 Requirements vs Testing Checks

Applicable: Yes

#	Check	Outcome	Justification
REQTEST-01	Every requirement under review is covered by at least one test	Pass	<code>dotnet reqstream --enforce (root-tags system, quality)</code> reports zero unsatisfied or orphaned requirements anywhere in the <code>CanvasNet-Fonts-* / CanvasNet-Lib-Fonts*</code> area (fresh run against same-day TRX results); all remaining unsatisfied requirements reported by the tool are unrelated Quality/Platform/OTS process requirements, outside this review-set.
REQTEST-02	Tests verify the behavior described in each requirement	Fail	While <code>fonts.yaml</code> 's own <code>tests:</code> list for <code>CanvasNet-Fonts-TtcSupport</code> is now correct, the companion verification-design document (<code>docs/verification/canvas-net/fonts.md</code> , also in scope) still describes the TtcSupport scenario as proven by two tests, one of which does not exist — the <i>documented</i> behavior-to-test mapping for this requirement is therefore still inaccurate even though the underlying requirements file and reqstream enforcement are now clean. See Finding F1 (remaining).

2.14 Code vs Design Documentation Checks

Applicable: No

Not applicable: the only source file under review is test infrastructure with no corresponding design document of its own.

3. Conclusion

3.1 Summary of Findings

#	Check	Finding
F1 (remaining)	DOC-01, DOC-02, REQTEST-02	The verification design document still cites a non-existent test for TTC support. <code>docs/verification/canvas-net/fonts.md</code> names <code>TrueTypeFont_RealTtcContainer_FaceZero_RendersGlyfGlyphOutlineAsVisibleInk</code> in its Test Environment section (“load the two faces of the locally-assembled <code>OpenSans-SourceSans3.ttc</code> ... fixture”), in its Acceptance Criteria (“all seven real-fixture integration tests in <code>TrueTypeFontRealFontIntegrationTests.cs</code> pass” — only 6 exist), and in the <code>CanvasNet-Fonts-TtcSupport</code> Test Scenarios entry (“Tests: <code>TrueTypeFont_RealTtcContainer_FaceOne_...</code>, <code>TrueTypeFont_RealTtcContainer_FaceZero_...</code>”). This test method does not exist anywhere in <code>test/</code> (confirmed by exhaustive grep across the whole test tree). This is a partial fix from the prior review: <code>docs/reqstream/canvas-net/fonts.yaml</code>’s own <code>CanvasNet-Fonts-TtcSupport.tests:</code> list was correctly updated to drop the phantom test (now citing only <code>TrueTypeFont_RealTtcContainer_FaceOne_RendersCffGlyphOutlineAsVisibleInk</code> and <code>CanvasNet_SystemIntegration_LoadFaceFromTtcContainerByIndex_ReturnsExpectedFace</code> both of which exist and pass), and <code>dotnet reqstream --enforce</code> now reports this requirement fully satisfied — but the verification document itself was not updated to match, and still misrepresents the actual test evidence for glyf-flavored face-0 TTC rendering. Confidence: High (directly verified via exhaustive grep of <code>test/</code>, confirmed the file has exactly 6 test methods not 7, and cross-checked against the now-corrected <code>fonts.yaml</code>).
F2 (resolved)	—	Previously: a cited test for TTC support did not exist in <code>fonts.yaml</code>’s own <code>tests:</code> list, causing <code>CanvasNet-Lib-FontsTtcSupport / CanvasNet-Fonts-TtcSupport / CanvasNet-Fonts-TrueTypeFont-LoadTtcExplicitFaceIndex</code> to be reported unsatisfied. Confirmed fixed: <code>fonts.yaml</code> no longer cites the phantom test; fresh <code>reqstream --enforce / --matrix</code> shows these requirements fully satisfied (1/1/0/0, no orphans).
F3 (resolved)	—	Previously: four unit-level CFF/Type1 requirements (<code>CffSeacComposition</code> , <code>CffCharsetResolution</code> , <code>CffWidthResolution</code> , <code>LoadType1C</code>) were orphaned, missing from <code>CanvasNet-Fonts-CffOutlineDecoding</code>’s <code>children:</code> list. Confirmed fixed: all four now appear in the <code>children:</code> list; fresh <code>reqstream --enforce</code> reports zero orphaned requirements in the Fonts area.
F4 (resolved)	—	Previously: the parent system-level design doc (<code>docs/design/canvas-net.md</code> , context) omitted <code>SystemFontCatalog</code> entirely. Confirmed fixed: it now fully documents <code>SystemFontCatalog</code> as the subsystem’s second public unit, with dependencies and an external-interfaces table entry for each of <code>Fonts</code> , <code>FindBestMatch</code> , and <code>LoadBundledFallback</code> , consistent with the in-scope subsystem design doc.
F5 (resolved)	—	Previously: <code>.reviewmark.yaml</code>’s <code>needs-review</code> patterns silently excluded <code>BundledFonts/README.md</code> (and <code>.csproj</code> files) from <code>--elaborate / --enforce</code> . Confirmed fixed: <code>needs-</code>

#	Check	Finding
		<code>review</code> now includes <code>**/*.csproj</code> and <code>**/README.md</code> patterns; <code>dotnet reviewmark --elaborate CanvasNet-Fonts</code> now correctly lists all 5 configured files, including the README.
F6 (partially resolved, out of scope)	—	Previously: <code>dotnet reviewmark --plan --enforce</code> reported 3 Fonts-area files globally uncovered by any review-set. Partially fixed: <code>src/DemaConsulting.CanvasNet/Fonts/CffStandardEncoding.cs</code> is no longer listed (now covered), but <code>test/DemaConsulting.CanvasNet.Tests/Fonts/BundledNotoFontsTests.cs</code> and <code>test/DemaConsulting.CanvasNet.Tests/Fonts/CffStandardEncodingTests.cs</code> remain globally uncovered (confirmed via a fresh <code>--plan --enforce</code> run, which now reports 15 uncovered files repository-wide, including these two). These are unit-level test files, correctly out of scope for this subsystem-level review-set; noted for the orchestrator's awareness only, not a finding against this review-set itself.
F-license (re-affirmed)	DOC-03 (Pass)	No change: <code>BundledFonts/README.md</code> 's licensing/provenance claims were re-checked (content unchanged since the prior review; the <code>BundledFonts/</code> folder itself was not touched by this documentation-sync pass) and remain accurate. No new licensing finding.

3.2 Overall Outcome

Overall Outcome: Fail

This re-review confirms that **4 of the 5 substantive findings from the prior review have been fully resolved** (F2 phantom test in requirements, F3 orphaned unit requirements, F4 stale parent design doc, F5 ReviewMark `needs-review` configuration gap), and the bundled font licensing/provenance documentation remains accurate. This represents clear, verifiable progress.

However, **one finding remains open and still causes an overall Fail:** `docs/verification/canvas-net/fonts.md` — itself a file under review in this review-set — still cites the non-existent test `TrueTypeFont_RealTtcContainer_FaceZero_RendersGlyfGlyphOutlineAsVisibleInk` in three places (Test Environment, Acceptance Criteria's test count, and the `CanvasNet-Fonts-TtcSupport` Test Scenarios entry), even though the companion requirements file (`fonts.yaml`) was correctly fixed to no longer cite it. The documentation-sync pass fixed the machine-enforced artifact (`reqstream` 's `tests:` list) but missed the parallel human-readable verification design document that makes the identical claim.

Required follow-up action before approval:

- Update `docs/verification/canvas-net/fonts.md` to remove the three references to `TrueTypeFont_RealTtcContainer_FaceZero_RendersGlyfGlyphOutlineAsVisibleInk :`
 - In the Test Environment section's fixture bullet list, describe only the one real `FaceOne` TTC test that exists (or, if a face-0 rendering test is genuinely intended to exist, implement it and then restore the reference once the test is real).
 - Correct the Acceptance Criteria's test count from "seven" to the actual count of tests in `TrueTypeFontRealFontIntegrationTests.cs` (6).
 - Correct the `CanvasNet-Fonts-TtcSupport` Test Scenarios entry's **Tests** line to match `fonts.yaml` 's own (now-correct) `tests:` list for that requirement
(`TrueTypeFont_RealTtcContainer_FaceOne_RendersCffGlyphOutlineAsVisibleInk ,`
`CanvasNet_SystemIntegration_LoadFaceFromTtcContainerByIndex_ReturnsExpectedFace`).
- Re-run this review after the above correction is made.

No corrections were made by this review; all findings are reported for the orchestrator to act on, per this review's read-only audit mandate.